INCH-POUND

A-A-54833
19 April 1993

COMMERCIAL ITEM DESCRIPTION

DRILL, HAND, BONE, Smedberg

The General Services Administration has authorized the use of this Commercial Item Description as a replacement for Military Specification, MIL-D-37215 which is cancelled.

This Commercial Item Description covers a bone drill brace, universal, hand type, of the Smedberg design, suitable for use in orthopedic surgery.

Salient characteristics:

Shall be a bone drill brace, universal, hand type, of the Smedberg design, suitable for use in orthopedic surgery. The bone drill brace shall have a one-piece body with removable gear container cover, No. "0" Jacobs type chuck, pistolgrip formed handle, Allen Wrench type handle chuck key, and thumb operated locking lever. Assembly of chuck to drill shank shall be by means of suitable mating tapers as indicated in Figure 1 of this specification.

The driving mechanism shall consist of two bevel gears; one secured to the drill shank and the other gear secured to the operating handle. The gears shall be designed in such fashion that one complete revolution of the operating handle shall result in two complete revolutions of the chuck. The thumb locking lever shall be of the cam-locking design that shall prevent turning of the operating handle when the locking lever is not depressed. The entire assembly shall be suitable for use and accept Steinmann type bone pins, bayonet and trocar points, triangular shank, sizes 3/32 to 5/32 inch in diameter; 4 to 12 inches in overall length; bone twist drills sizes 3/32 to 5/32 inch in diameter; and Kirschner bone pins sizes 0.035 to 0.062 inch in diameter by 9 to 12 inches in overall length. The drill assembly shall be cannulated throughout its length to accept drills up to 12 inches in overall length.

AMSC N/A FSC 6515

DISTRIBUTION STATEMENT A: Approved for public release; distribution is unlimited.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T09:39Z
Check the source to verify that this is the current version before use.

The No. "0" Jacobs chuck shall have a No. "0" taper, as indicated in Figure 1, with unthreaded tapered hole. The chuck shall accept bone pins and drills having diameters up to, and including, 5/32 inch. The chuck key shall be of the universal type suitable for use with the No. "0" Jacobs chuck. The key handle shall be of the Allen Wrench Type suitable for use with number 10 socket head screws. The key gear shall mesh easily with the chuck gear without slipping. The complete drill assembly shall be in accordance with Figure 1. As a alternate, the key handle shall be Allen Wrench type only on the outermost (0.156 ± 0.010 inches) of both ends.

Material. The drill body shall be of a suitable aluminum alloy. The drill shank shall be of stainless steel Class 3 or 4 in accordance with ASTM F899, Rockwell Hardness B70 to B95 (for Class 3) and C36 to C41 (for Class 4). The operating handle shall be of brass, or copper alloy.

The Jacobs Chuck shall be of Class 4, stainless steel in accordance with ASTM F899, Rockwell Hardness C36 to C41.

The thumb locking lever shall be made of nickel plated brass, Class 4 stainless steel in accordance with ASTM F899, or copper alloy.

The Chuck Key shall be Class 4 stainless steel in accordance with ASTM F899, Rockwell hardness C36 to C41, or Type SAE 1119 or 1117 carbon steel, Rockwell Hardness C50 to C60.

The Chuck Key gear shall be Class 4 stainless steel in accordance with ASTM F899, Rockwell Hardness C36 to C41. Or Type SAE 1119 or 1117, Rockwell Hardness C50 to C60.

The Chuck Key handle shall be of Class 4 stainless steel in accordance with ASTM F899, Rockwell Hardness C36 to C41, or Type SAE 1119 or 1117 carbon steel, Rockwell Hardness C50 to C60.

As an alternate, the drill shank and Jacobs chuck may be made of 440F, providing this material complies with corrosion resistance requirements for Class 4 material of ASTM F899.

Style, design and dimensions. The bone drill shall be in accordance with Figure 1 for style, design and dimensions.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T09:39Z
Check the source to verify that this is the current version before use.

Finish. Finish of the drill body, Jacobs chuck and chuck key shall be mirror, or mirror and satin.

Passivation. All material made from Class 3 or 4 stainless steel shall be properly passivated.

Identification marking and labeling. The bone drill shall be legibly marked with the contractor's name of trademark and the word "stainless". Foreign manufactured drills shall be legibly marked with the name of the country of origin.

Workmanship. The bone drill shall be free from defects which detract from its appearance or impair its serviceability.

Unit. Each (EA). One drill hand bone or one chuck drill or one key drill, as specified, constitutes one unit. Complete and detailed instructions for use shall be supplied with each unit.

QUALITY ASSURANCE PROVISIONS.

Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

Records. Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government upon the Government's request, at any time, or from time to time, during the performance of the contract and for a period of three years after delivery of the supplies to which such records relate.

Inspection. Inspection, as used herein, is defined as both examination (such as visual or auditory investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties) of the item.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T09:39Z
Check the source to verify that this is the current version before use.

Examination. Examination shall be conducted to determine compliance with specification requirements.

Sampling for examination. Sampling for examination shall be conducted in accordance with MIL-STD-105 and as specified. Unit of product for examination purposes shall be one complete bone drill assembly or one chuck, or one key as specified. Sampling shall be inspection level II with an Acceptable Quality Level (AQL) of 1.0.

Tests. Tests shall be conducted to determine compliance with specification requirements. Where feasible, the same sample shall be used for determination of two or more test characteristics.

Sampling for tests. Sampling for tests shall be conducted in accordance with MIL-STD-105 and as specified. Unit of product for test purposes shall be one complete bone drill assembly, or one chuck, or one key. Sampling shall be inspection level S-2 with an AQL of 1.0.

Contractor certification. The contractor shall certify and maintain substantiating evidence that the product offered meets the salient characteristics of this Commercial Item Description, and that the product conforms to the producers' own drawings, specifications, standards, and quality assurance practices. The government reserves the right to require proof of such conformance prior to first delivery and thereafter as may be otherwise provided for under the provisions of the contract.

Metric products. Products manufactured to metric dimensions will be considered on an equal basis with those manufactured using inch-pound units, providing they fall within the tolerance specified using conversion tables contained in the latest revision of Federal Standard No. 376, and all other requirements of this document are met.

If a product is manufactured to metric dimensions and those dimensions exceed the tolerances specified in the inch-pound units, a request should be made to the contracting officer to determine if the product is acceptable.

The contracting officer has the option of accepting or rejecting the product.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T09:39Z
Check the source to verify that this is the current version before use.

Regulatory requirements.

Federal Food, Drug, and Cosmetic Act. If the product covered by this document has been determined by the U.S. Food and Drug Administration to be under its jurisdiction, the offeror/contractor shall comply, and be responsible for compliance by its subcontractors/suppliers, with the requirements of the Federal Food, Drug and Cosmetic Act, as amended, and regulations promulgated thereunder. In addition, the offeror/contractor shall comply, and be responsible for compliance by its subcontractors/suppliers, with the requirements of all other applicable Federal, State, and local statutes, ordinances, and regulations.

Recovered materials. The offeror/contractor is encouraged to use recovered material in accordance with Federal Acquisition Regulation Subpart 23.4 to the maximum extent practical.

Preservation, packaging, packing, labeling, and marking. Unless otherwise specified, preservation, packaging, and packing shall be to a degree of protection to preclude damage to containers and/or contents thereof under normal shipping conditions, handling, etc., involving shipment from the supply source to the receiving activity, plus reshipment from receiving activity and shall conform to applicable carrier's rules and regulations. Intermediate and exterior package quantities and labeling and marking shall be as specified in the contract and/or order.

Ordering data. Intermediate/exterior package quantities and labeling and marking must be specified in the contract and/or order.

NOTE: The following National Stock Number and its respective Item Identification is covered by this document:

NATIONAL STOCK NUMBER	ITEM IDENTIFICATION
6515-00-312-3500	DRILL, HAND, BONE, Smedberg

MILITARY INTERESTS:

Custodians:

Army - MD
Navy - MS
Air Force - 03

Preparing Activity:

DoD-MB

Agent:

DLA-DM

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T09:39Z
Check the source to verify that this is the current version before use.

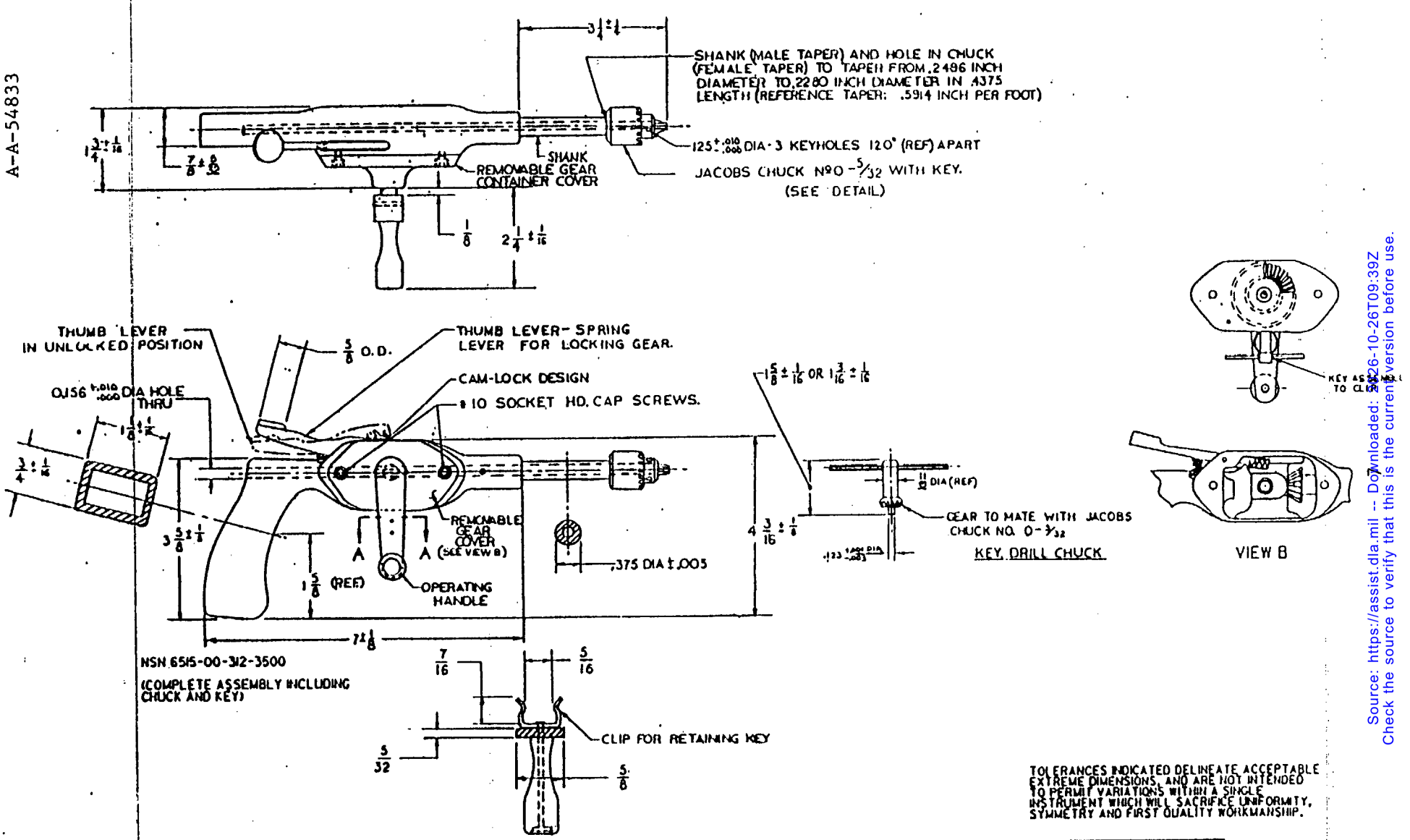
CIVIL AGENCY COORDINATING ACTIVITIES:

VA-OSS
USPHS
FDA-MPQAS

Project Number: 6515-5432
Location: c:\eric\drillcid

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T09:39Z
Check the source to verify that this is the current version before use.

A-A-54833



TOLERANCES INDICATED DELINEATE ACCEPTABLE EXTREME DIMENSIONS, AND ARE NOT INTENDED TO PERMIT VARIATIONS WITHIN A SINGLE INSTRUMENT WHICH WILL SACRIFICE UNIFORMITY, SYMMETRY AND FIRST QUALITY WORKMANSHIP.

UNLESS OTHERWISE SPECIFIED DIMENSIONS ARE IN INCHES.		
TOLERANCES ON		
FRACTIONS	DECIMALS	ANGLES
± 1/32	±	±

FIGURE 1 DRILL, HAND, BONE, SMEDBERG

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T09:39Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8 In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER	2. DOCUMENT DATE (YYMMDD)
	A-A-54833	19 April 1993

3. DOCUMENT TITLE
DRILL, HAND, BONE, Smedberg

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)*

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME	b. TELEPHONE (Include Area Code) (1) Commercial	(2) AUTOVON
Defense Personnel Support Center ATTN: DPSC-MSE	(215)7372870	444-2870
c. ADDRESS (Include Zip Code) 2800 South 20th Street Philadelphia, PA 19145	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340	

Source: https://assist.dla.mil - Downloaded: 2026-10-26T09:39Z
Check the source to verify that this is the current version before use.